# Room for the River

Identifying model-based strategies for the Transport Company

EPA141A

Group 1

*(HNS, n.d.)*


**Authors:**

Daniela Ríos Mora, No. 6275486

Amaryllis Brosens, No. 5307554

Celia Martínez Sillero, No.6222102

Oscar Rajantie Tuñon, No. 5477913

TU Delft

Faculty of TPM


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# Summary

The *Room for the River* project is a major national effort to improve flood safety in the Netherlands by redesigning rivers and their surroundings. It is a large-scale, long-term intervention in the IJssel river that affects both the natural environment and the people who live and work in flood-prone areas. The *Room for the River* project is subject to deep uncertainty. This means parties cannot agree on the system, its boundaries, the probability of uncertain inputs, and/or the outcomes of interest (Lempert et al., 2003; Walker et al., 2013; Kwakkel et al., 2016b). In addition to this uncertainty, the project is embedded in a complex network of stakeholders, including national ministries, regional water boards, municipalities, local residents, environmental groups, private landowners and companies. These stakeholders often have conflicting interests and different perspectives on what the problem is and how it should be addressed. Some prioritize safety, others economic development, heritage preservation, or ecological values. The situation is dynamic since views and priorities can shift over time, making it even more challenging to identify solutions that are stable, widely supported, and future-proof.

The present report focuses on the perspective of the Transport Company (TC), whose main objective is ensuring the Minimum Water Level (MWL) of 4 meters throughout all points of the IJssel River.

Given this complexity and uncertainty, this analysis focuses on identifying robust policies. Rather than identifying a single "optimal" solution, we explore policies that perform reasonably well across a wide range of plausible futures. This approach helps test how different policies might hold up under changing assumptions and scenarios, and which strategies are most resilient to surprise or disagreement (Kwakkel et al., 2016b).

As a result, the report suggests two robust policies that ensure the TC's main goal while minimizing the *Expected Annual Deaths* and *Expected Annual Damage*. The latter two are interests foreign to the TC that were added to ensure alliances during the political debate. This was decided after addressing that the TC lacks political relevance to force a specific policy and that most cases lead to acceptable MWL to TC, in contrast to initial concerns. Hence, TC should just avoid the worst policies and strengthen its relevance by making alliances with the higher authorities like Delta Commission and Rijkswaterstaat.

Lastly, the proposed policies for the TC to advocate for are policy 2 and 9. Both of them consist of two rooms for the river projects in Gelderland, (Olburgen and Tichelbeeksewaard), two dike increases in Gelderland: Cortenoever and Zutphen, and two dike increase locations in Overijssel: Gorssel and Deventer. Policy 9 adds a *Room for the River* project Obstakelverwijdering.

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# 1. Problem Framing

The Netherlands need to develop a comprehensive flood protection strategy due to an increased focus on the concept of resilience in flood protection. This has also called into question the actual effectiveness of the traditional protection policy paradigm (dike building and heightening). River and flood management requires a system perspective, whereby policy analysts and policymakers should consider the entire river and its functions, rather than just regional sections or the flood protection function. A systems perspective also involves considering a large number of feedback and feed-forward mechanisms between the subsystems. The prevention policies chosen will affect the water level, which then impacts how the river is used (such as for farming or transporting goods). These uses, in turn, influence the water level again (De Bruijn et al., 2015). Additionally, when dealing with flood risk management, it is not only a matter of safety; the local, regional and national economies must also be considered. This makes flood risk management in the Netherlands a potential multi-issue game where multiple stakeholders can be involved, as their interests may easily become intertwined with any intervention planned by political and water authorities. Any major change to the Dutch hydrographic structure and flood disaster prevention measures will have consequences for the short, medium and long term. This means that policy-makers cannot afford to make a mistake (Kwakkel et al., 2016b).

Unlike previous flood risk strategies, *Room for the River* (RfR) project opens the solution space for other policies than the traditional dike building and its periodic heightening (De Bruijn et al., 2015). The political decision-making process behind the RfR project is inherently complex. To assist the Transport Company on the decision-making process, this report will analyze the causes of this complexity. According to the academic literature, complexity in projects comes from problems being multi-layered, and being subject to deep uncertainty (Van Ernst et al., 2014; Rittel & Webber, 1973).

The first one refers to problems involving an array of stakeholders with conflicting stakes and needs. Such a tangled and dense network of interests tends to result in every problem formulation being contested. Why would such an initial step in the decision-making process be already questioned? Precisely because it is the first step. Problem formulation defines the first boundaries of the solution space. Hence, if the question does not look or cover your interests, it most likely won't lead to a desirable outcome for you as a stakeholder.

Secondly, deep uncertainty is a term to describe a higher number of unknowns and ignorance when tackling a grand challenge. Beyond "risk", policy also deals with 'unmeasurable uncertainty', meaning that many times there are unknown unknowns (Stirling, 2010).

Overall, the RfR project checks both boxes, meaning that it can be considered a ‘wicked problem’. Unlike "tame" or "structured" problems that may have definitive solutions, wicked problems have no definitive formulation or solution, and their resolution depends heavily on the values and interests of the many stakeholders involved (Rittel & Webber, 1973). They are systems of problems rather than isolated ones (Ackoff, 1979). More information often worsens these problems by providing more material for differing positions.

As there is scientific uncertainty and societal conflict, many problem formulations can arise. In this report, the case of the Transport Company (TC) will be studied in further detail from an analyst perspective. Nevertheless, it is crucial to address other framings to get an understanding of the multi-actor environment. This will allow for better negotiations practices based on the model since proposals will count other stakeholders’s interests.

According to the problem owner of this report, the Transport Company, the Dutch water authorities are considering alternatives to traditional flood risk management policies (e.g. increasing dikes’ height). The government might be pressured by the downstream provinces (Province of Overijssel, Dike Ring 4 and Dike Ring 5) to reduce the volume of water flowing in the river. Reducing the volume of water flow in the river is not interesting for the TC since it will affect the economic feasibility of its business model. TC problem formulation will be:

***“How can the TC be relevant in the IJssel river project decision-making process without compromising the minimum water level?”***

Actors like Delta Commission may see this project differently. Their views could be considered competitive since both stakeholders have initially opposing goals. They see the project as crucial to enhance long-term safety for the population in the Netherlands. Hence, their framing could be: "*How can Delta Commission minimize the number of deaths and expected annual damage in the RfR project without incurring unreasonable costs?*"

To further understand the impact of different perspectives in the *Room for the River* project, we now compare the Province of Overijssel and Gorssel’s possible problem formulation.

On the one hand, the Province of Overijssel must achieve consensus in its region as much as possible. This means achieving a plausible outcome for its dike rings, cities and villages. The main goal of this actor is reelection, which may imply in worse scenarios to at least make as many people as possible happy from the region. This aligns with a utilitarian perspective, but probably would end up focusing on the areas with largest population concentration, the main cities.

On the other hand, the village of Gorssel sees the possible change of paradigm in flood protection as a threat. This is a rural area that relies heavily on agriculture, any measure related to Room for the River policies would lead to giving up land, which could critically harm their economy. Nevertheless, their reliance on agriculture also makes them aware of the flood risks. According to our conversation with them, it seems they were willing at some point to contribute economically even, but not land-wise. Therefore, a potential framing from Gorssel could be: *how can Gorssel be best protected from flood by the Room for the River Project without hosting any intervention in its area?*

Overall, these research questions highlight how it is essential to establish partnerships with actors who possess decision-making power, whether through economic resources or other forms of leverage. A key concern is how the TC can best leverage its interests and develop an adaptive strategy that maximizes its chances of success. This question will be explored in the following sections.

# 2. Approach

A simulation model is developed to assess the feasibility of the Transport Company's objectives in the negotiation. This model uses the **XLRM** framework for its structure. **X** stands for the external factors that are out of control for decision-makers. **L** for Levers, these are the possible actions decision makers can take to leverage the Relationships in the system (**R**) to get desirable values in Measures (**M**), the outputs of the model (Kwakkel, 2017).

As stated before, the *Room for the River* project is subject to deep uncertainty and decision-makers can ill-afford to be wrong. (Lempert et al., 2003; Walker et al., 2013; Kwakkel et al., 2016b) These two factors call for robust policies that can succeed under different scenarios. Hence, this report will base its analysis aiming for robustness of the policies. It will use exploratory modelling, and more specifically the Exploratory Modelling Approach (EMA) to take into account the impact of the assumptions in the model and try different ones to test robustness. EMA allows the systematic exploration of uncertainties and its performance across many scenarios (Kwakkel et al., 2016a).

## 2.1 Open exploration

The client, the Transport Company, has a main objective : (at least) preserving the *Minimum Water Level* (MWL). In this analysis, the possible policies are evaluated prioritizing this outcome. First the scenario space is studied. In this part, the uncertain parameters (Appendix A) that lead to undesirable outcomes are identified. Then, in the policy space exploration, policies are evaluated without varying uncertainties. These two analyses provide a big picture of the input variables' range (**X** and **L**).

### 2.1.1. Scenario space

In this section, scenario discovery is applied to identify the uncertainty space. As the aim is to maximize the MWL throughout the system, the focus is on identifying scenarios that could lead to an undesirable low minimum water level.

First, the BAU policy is studied varying the uncertain inputs. This is done by doing a global sensitivity analysis. The goal is to understand which uncertain variables affect the most to the outcomes when there is no policy intervention. By addressing where the uncertainty comes from at the start of the analysis, the future steps will focus on policies that are robust especially against these critical uncertainties. To do so, the analysis employs first a Random Trees technique (RF). Based on Jaxa-Rozen and Kwakkel (2018), tree-based ensemble methods provide a high performance for screening influential factors, which is very suitable if interactions exist. This allows us to identify , by calculating GINI scores, the most influential uncertainties on the outputs. Moreover, it is suitable as a prior step to a SOBOL analysis. Since the latter is highly computationally expensive, RT can reduce its requirements by prioritizing the most relevant uncertainties based on the featuring scores.

SOBOL analysis quantifies the influence of uncertainties for a targeted scenario design. SOBOL provides a ranking of the factors including their main and interaction effects (Kwakkel et al., 2016a). After identifying the most influential uncertainties using Sobol analysis, those variables can inform the construction of “extreme” or “stress-test” scenarios, such as a worst-case scenario where outcomes (e.g., Minimum Water Level) are critically low. This scenario will serve as a reference scenario in the (many-objective) robust optimization stages.

Later on, the policy of do nothing (BAU) is tested using Scenario Discovery techniques. In this part, the aim is to identify which scenarios in the scenario space lead to undesired outcomes. To create the reference scenario, the results from SOBOL are used as input for the Primary Rule Induction Method (PRIM) algorithm. The PRIM is a scenario discovery technique that is performed on the outcomes of the filtered SOBOL. PRIM identifies subspaces of the input space that correspond to a target. The target in this case is a MWL below 5m. PRIM iteratively “peels” away parts of the data to find a region with a relatively high concentration of target outcomes (Bryant & Lempert, 2010). In PRIM the goal is to find a desirable trade-off in the uncertainty space between density, coverage and number of restricted dimensions by choosing a box with more than 0.6 coverage, 0.7 density, and just two restricted dimensions. Thereby, the algorithm’s imposed restrictions can cover most of the cases of interest while maintaining easy interpretability. A subspace or box with medium coverage and high density is chosen. The PRIM box boundaries with their midpoints are used to create the reference scenario.

### 2.1.2 Policy space

Once the system’s input uncertainty is explored, random policies are tested across various futures to map the general performance landscape. EMA multiprocessing evaluator is applied with random policies. This means that the algorithm samples random combinations of levers and tests them across different random scenarios, which allows discovering sets of actions that are worth further inspecting (Kwakkel et al., 2016a). In this case, the latin hypercube with uniform distribution is used as a sampling technique. This allows us to address all the policy space regardless of their popularity in the debate. This is done at an early stage of the analysis to get a general idea of the possible look of the outcomes with a neutral disposition. Only the variable of *A.0_ID flood wave shape* is made constant to 30, like in the reference scenario. This is because flood wave shape is a categorical variable with no inherent ranking and this could add noise into the analysis without adding meaningful insight. The flood wave shape will be explored in later experiments of robust decision-making.

After exploring the lever space randomly and gathering a general view of the system without favoring any policy, specific lever combinations need to be examined. These selected policies are made based on information gathered during the first debate and on requests from the transport company (Appendix B). The PRIM technique is used in these experiments to determine the scenario space in which policies could be favourable for the Transport Company.

## 2.2 Direct Search

One of the most common use cases for direct search is to find the values for the policy levers that maximize desirable outcomes. As previously mentioned, our client's primary goal is to preserve the MWL, and Many-Objective Robust Decision Making (MORDM) provides a framework to generate robust candidate solutions when dealing with deep uncertainty (Eker & Kwakkel, 2018). A robust solution is one that performs well across a range of possible future states of a system (Bartholomew & Kwakkel, 2020).

MORDM analysis is suitable for our client, as it enables the exploration of policies that can align with the interests of other actors while meeting its objectives. As it became clear during the debate exercise, the Transport Company holds no political power and needs strategic alliances to achieve its goals.

The framework consists of four phases: a problem formulation based on a systems analytical problem definition, the search for candidate solutions with Many-Objective Evolutionary Algorithm (MOEA), the analysis of trade-offs where strategies are compared across many objectives and a scenario discovery phase to detect vulnerabilities of the candidate solutions (Watson & Kasprzyk, 2017; Ekker & Kwakkel, 2018).

MORDM is based on a single reference scenario (Eker & Kwakkel, 2018). As analysts, we agreed to target the weakness of the single-scenario approach by searching for candidate strategies based on the worst-case scenario, which was obtained from the scenario discovery performed using PRIM.

The search for candidate solutions was repeated using five different seeds to reduce stochastic uncertainty, and the final results were merged into a set of Pareto-optimal strategies. Computational power was constrained, as every run lasted 2 hours on our computers, and access to the Delft Blue was provided; however, the job remained in the queue.

### 2.2.1 Problem Formulation

In the MORDM problem formulation, the metrics to be optimized are defined, along with the available policy levers to be tested. The metrics to be optimized include the MWL which is to be maximized, and the metrics to be minimized: Expected Annual Damage, Investment Costs, Evacuation Costs, and Expected Number of Deaths. The available policy levers are a set of policies with room for the river projects in locations Olburgen, Havikerwaard, Tichelbeeksewaard, Welsummer buitenwaarden, and Obstakelverwijdering (see appendix A). For the uncertain factors, the reference scenario is used as the initial values.

### 2.2.2 MOEA

The goal in an MOEA is to first find a set of well-distributed solutions close to the true Pareto-optimal front (Deb et al., 2005). A Pareto-optimal consists of several solutions that are not dominated by any other solution and that are distributed with a uniform density across the sampling space (Maier et al., 2019).

To find these solutions, the optimize function of the EMA workbench is used. This function uses the $\varepsilon$-NSGA-II algorithm to search for solutions based on their non-dominance and the distance to their neighbours. This algorithm utilises $\varepsilon$-dominance, which enables us as analysts to assign a relative importance to each objective, also known as the $\varepsilon$-value. It defines a grid of equally-sized blocks, and the $\varepsilon$-values determine the shape of the Pareto-front. A high $\varepsilon$-value results in a coarse grid, while a low $\varepsilon$-value results in a fine grid (Kollat and Reed, 2005). Subsequently, dominated solutions are not considered. The algorithm provides a set of candidate solutions based on the reference scenario. To evaluate whether the algorithm has reached convergence, the metrics Epsilon Progress, Epsilon Indicator, Generational Distance, and Inverted Generational Distance are used.

The optimizer is run with an *nfe* of 50000 to ensure convergence. Additionally, it is run with 5 different seeds to address stochastic uncertainty and getting stuck in local optimal, maximizing the chance to get a well-distributed pareto-front.

The candidate solutions are further reduced by selecting the policies that comprise the top 25% *Expected Number of Deaths* and *Expected Annual Damage* values. During the first debate, the province of Overijssel was keen to keep the number of deaths as low as possible. Later, during the second debate, the Delta Commission and Rijkswaterstaat stressed its importance, as they requested that the actors provide the numbers for the monetary and casualty aspects of their desired strategy in comparison with the proposed policy from Rijkswaterstaat.

### 2.2.3 Robustness Analysis

To evaluate the robustness of the candidate strategies, the *MultiprocessingEvaluator* and the *perform_experiments* method are utilised. It utilizes the Latin Hypercube sampler by default to evenly represent the uncertainties and policies, allowing for the assessment of the effect of uncertainties (Kasprzyk et al. 2013).

The chosen robustness metrics are signal-to-noise ratio and the maximum regret.

#### 2.2.3.1 Robustness Metrics

Maximum regret is used as a robustness metric as in (Eker & Kwakkel, 2018). A lower value maximum regret value is preferred. Since a MWLI value is used as a constraint in the policy search, the pareto optimal solutions satisfying the constraint ought to be highly performing solutions from the point-of-view of the TC (Piscopo et al., 2015).

Maximum regret is the highest difference between the worst-case scenario for a policy and the best performing scenario with the best performing policy. It is calculated from the aggregated values, only a single value – such as the highest expected number of casualties – is used to calculate the regret metric. The high level of aggregation means that more intricate details, such as the distribution of values, is not taken into account. Using maximum regret leads to conservative solutions since it is calculated from the worst performance in the scenarios (Bertsimas & Sim, 2004; McPhail et al., 2018). In other words, there is a high level of intrinsic risk aversion. Lower maximum regret values are preferred for all metrics.

In Eker & Kwakkel (2018), signal-to-noise ratio is also used as a robustness metric. It is a statistical metric of the ratio of mean over standard deviation. However, the signal-to-noise ratio has its downsides. First, its performance is highly model dependent and it is suitable for models with a linear relation between the mean and the standard deviation (Bérubé & Wu, 2000). Secondly, the signal-to-noise ratio does not take into account the direction of the deviation from the mean. Thus, good deviations (towards a preferred direction) affect the metric the same way as bad deviations (Takriti & Ahmed, 2004). Nevertheless, the signal-to-noise ratio has been used in Eker & Kwakkel (2018) and Hamarat et al. (2014) among other studies. Due to its use in the aforementioned studies, signal-to-noise ratio is used in this analysis despite its drawbacks. It represents the consistency of the policy performance by indicating the dispersion of scenario outcomes. Lower signal-to-noise ratios are preferred in all outcome variables.

An alternative approach would have been to use Starr's domain criterion (a satisfying metric) as the robustness metric. If the MWL is viewed as a hard constraint, a satisfying metric would have been appropriate (McPhail et al., 2018). However, in this case it is assumed that the water level depends on the vessel type and load, and thus is not a strict constraint but rather a continuum. Thus, the aim is to optimize system performance, making a regret-based metric suitable (McPhail et al., 2018). Looking at the criteria used in the shipping industry, the all-time minimum water level is not used as a hard criterion, rather a 95% or 98% guaranteed lowest navigable water level is used (Wang et al., 2021). Therefore, maximum regret can be used as an appropriate metric for this situation. In further research the Starr's domain criterion could be used to explore the solutions.

### 2.2.4 Scenario Discovery/PRIM

After the robust candidate strategies are selected, scenario discovery is performed with PRIM to identify in which scenarios they perform poorly (Eker & Kwakkel, 2018). As in the open exploration phase, one box is identified. The first one contains the cases in which the *MWL* is below 4.8.

The PRIM box boundaries are used to detect vulnerabilities in the candidate solutions.

### 2.2.5 Candidate Strategies

The final step of the MORDM framework is to select a final set of candidate strategies. These are chosen based on the robustness metrics scores. The worst policies are dropped and the best ones serve as a recommendation for the TC.

# 3. Results

The following section discusses the results obtained after applying the previously described approach.  While this section aims to be a summary of the results, further information can be found in the appendix if desired.

As a reminder, in this state of the analysis, the main objective of the Transport Company (TC), is to (at least) not decrease the current MWL with the new policies and projects. Therefore, as analysts, this report aims to identify policy combinations that avoid worsening the MWL. These policies should be robust across scenarios and acceptable to other stakeholders. This allows TC to negotiate preferred policies with them.

## 3.1. Open exploration

### 3.1.1. Scenario space

The Random Trees (RT) algorithm points out that the A.0._ID flood wave shape is the most influential variable when determining the outcomes. On the other hand, the rest of the uncertainty variables seem to have the same order of magnitude). This means that RT does not provide a strict prioritization and all variables influence somehow the same (Appendix B.1).

Therefore, SOBOL analysis is done taking into account all variables, except flood wave shape. This is because flood wave shape is a categorical variable and SOBOL sampling technique can only use continuous variables. Based on the RT, flood wave shape will be studied with more detail in the Many-Objective Robust Decision Making. For the sake of continuing the Open Exploration, the flood wave's shape is set to 30 arbitrarily.

SOBOL analysis results show that AA.1_B.max, A.1_p_fail, A.2_Bmax, A.2_pfail, A.4_pfail and A.3_pfail have the most impact on MWL. Figure 1 displays the results for MWL. The blue bars quantify the first order effect on the outcome per variable. The orange bars quantify the total effect, the first-order and the second order effect.  The second-order effect evaluates how much specific interaction between said variable and other variables affects MWL.  It is noteworthy to have stronger second-order interactions than first-order. This means that variables impact most outcomes when interacting with other variables rather than on their own, which makes sense because MWL continually depends on the whole river.

On the other hand, discount rates do not seem to affect.

Lastly,the A.5 dike ring does not play a big role. This makes sense as it is the most downstream point, closest to the coast. If something happens there, it is close enough to the sea to not affect the whole flow of the Ijssel river.

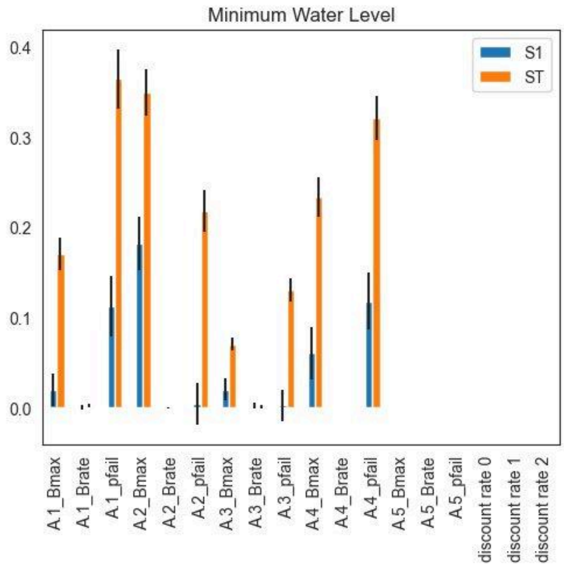


Figure 1. SOBOL Results for MWL

Then, SOBOL analysis' results are used for PRIM. Aiming for a reasonable trade-off between density, coverage and number of restricted dimensions, a box is chosen for further study. The algorithm converges into two constraints: Chance of failure in Doesburg dike ring (A1) higher than 34% and breach width in Cortenoever dike ring (A2) is higher than 140 meters. These conditions cover 69% of the situations where MWL is less than 5 meters. If these conditions are met, there is a chance of 73% that MWL is undesired. These conditions are used to develop a worse-case scenario where MWL is not met when there is no policy. This reference scenario will be used in the MORDM subsection. The specific scenario can be found in the Appendix B.2. It has relatively low discount rates, rather high Bmax values and average pfails. Additionally, the flood wave shape is fixed to 30.

### 3.1.2. Policy space

#### Random policies

The sensitivity analysis of the exploration of the random policies highlight the uncertainties that are most influential to the outcomes, the results can be found in Appendix B.3.

For the MWL, the most influential policy lever is room for the river, particularly 3_RfR. This is the RfR project at Welsummer Buitenwaarden. In addition, dike increase measures also affect minimal water level. The influence is spread over the different dike rings, with the most influential one being the dike increase in dike ring

1. This is consistent with expectations as this area is located upstream and can influence the water level downstream.

The dike increase in ring 1 is also the most influencing for the outcome expected annual damage (EAD). For EAD, other influential levers include the dike increase in ring 3 (Zutphen), which is an urban area and 2_RfR, which is also in the area of Zutphen.

Regarding the expected number of deaths, the most impactful lever is the dike increase in A5, which corresponds to Deventer. This finding aligns with expectations, as Deventer is a densely populated urban area, making the potential impact more severe in terms of human lives.

#### Selected policies

The PRIM of the selected policies (Appendix B.4), targeting a MWL below 5m, shows that the only restricting parameter is the policy choice. This means that the outcome of the system mostly depends on the chosen policy rather than by uncertainties. This results aligns with our expectations because in the BAU scenario the MWL generally stays above 5m.

While we cannot draw any definite conclusions, it can be observed that there is a tendency that policies with more *Room for the River* locations are more often associated with *MWL* below 5m. However, no clear pattern emerges regarding whether upstream or downstream RfR have more influence. The timing of implementation also appears to have limited impact. In some cases, dike heightening seems to mitigate the effects of RfR.

Finally, resampling confirms that the policy variable can reproduce coverage and density, indicating that the box's performance is robust and not due to overfitting.

## 3.2. Direct Search

The optimization of multiple seeds is based on the reference scenario presented in Appendix B.2.

### 3.2.1 MOEA

In total, five different seeds are used to control stochasticity. For every seed, the $\varepsilon$-progress was almost stabilized. The generational distance, the ε-indicator and the spacing metrics stabilized, meaning that it stopped improving significantly in terms of proximity to the true Pareto front, the distribution is even and that it is not improving considerably in comparison to the reference scenario. It is reasonable to consider that it converged, but a higher number of function evaluations (*nfe*) or a greater variety of $\varepsilon$-values might slightly improve the outcomes. Due to computational power constraints, we decided to stick to these results.

After merging the candidate solutions from the different seeds, an overall pareto optimal was achieved using $\varepsilon$-dominance (epsilon_nondominated( ) in EMA workbench). An ε-dominant solution is one that is no worse than another solution by more than ε in any objective. Epsilon-non-domination is a filtering function to merge results into a single set of nondominated solutions. It keeps the solutions which are not ε-dominated by others. Since a tolerance value of ε is used, it is more tolerant compared to a Pareto optimality which would filter the solutions without a tolerance value. As ε is larger, the set of solutions is larger with solutions further from Pareto optimality. When $\varepsilon=0$, ε-non-domination function would give the Pareto optimal solutions. In this search, a value of $\varepsilon=0.05$ is used.

The use of five different seeds reduces the impact of stochasticity and partially compensates for the smaller nfe. Merging the seeds with ε-non-domination increases the effective number of scenarios, thus improving robustness and coverage of the solution space.

The Pareto optimal solutions were reduced using the numpy percentile function since all the candidate strategies reached a *MWL* above 4. We obtained the 25% of outcomes with lowest *Expected Number of Deaths* and *Expected Annual Damage.* Thereafter, the results with the 10 best *MWL* were chosen.

From the Trade-offs between outcomes shown in picture 2 we can infer the next:

- All candidate policies have a desirable value of *MWL*.
- A higher *MWL* leads to an increase in the *Expected Number of Deaths*; however, the resulting values are not significantly high, meaning it does not translate into a large number of casualties.
- Policies that have a higher *Dike Investment Costs* incurred in lower *RfR Investment Costs.*
- There is a clear trade-off between *MWL* and *Expected Annual Damage.*

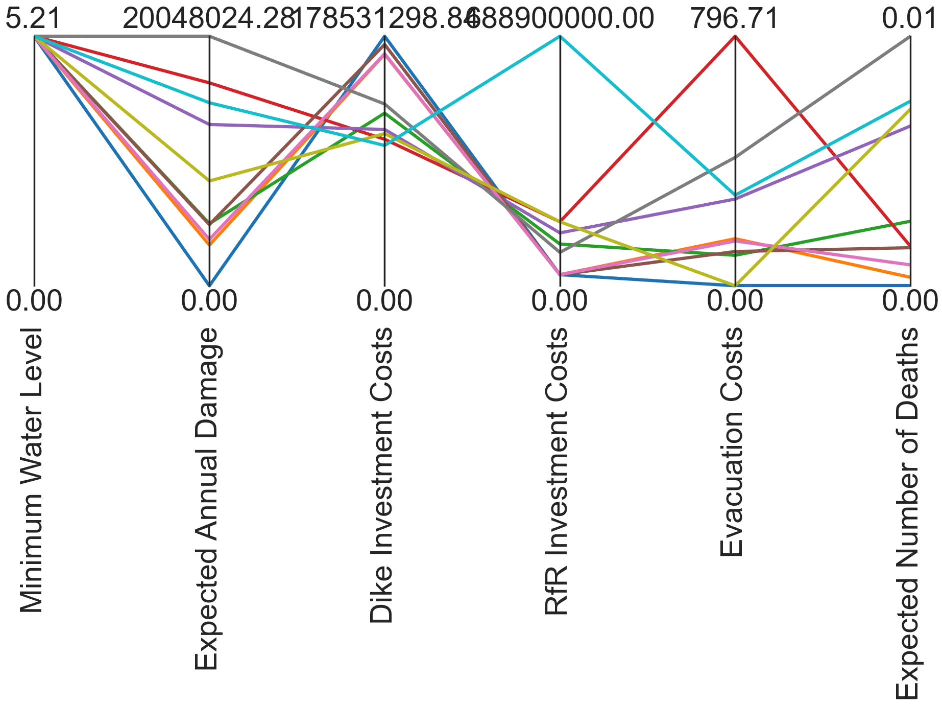


*Figure 2: Trade-Offs between outcomes for 10 candidate policies*

Eight different *A.0_ID flood* values in the reference scenario were used to explore potential sensitivity to this uncertainty parameter in the outcomes since it could not be analyzed in previous stages of the report. Variations in this parameter did not result in any scenario with a MWL below 4 meters. Furthermore, the plots in Appendix C suggest that the trade-offs behave similar across simulations. For instance, a higher MWL incurs an increase in the *Expected Number of Deaths.* However, there is a difference in the outcomes values and the candidate policies.

Eight different A.0_ID flood values in the reference scenario were used to explore the sensitivity of outcomes to this uncertainty parameter, which had not been analyzed in earlier stages of the report. Variations in this parameter did not result in any scenario with a MWL below 4 meters. Moreover, the plots in Appendix C indicate that trade-offs behave similarly across simulations. For example, a higher MWL generally leads to an increase in the *Expected Number of Deaths*. However, there are differences in the outcomes values and the selected candidate policies. No definitive conclusions can be drawn about the sensitivity of outcomes to this parameter.

### 3.3.2 Robustness

The first robustness metric used is the signal-to-noise ratio. The most important outcome variable is the MWL. As can be seen from Figure 3, policies 9 and 2 are the best and second-best performing policies on this metric, respectively.

The other policies all have normalized signal-to-noise ratio values of 0.4 or higher. Policy 9 has an absolute signal-to-noise ratio value of 12.8 and policy 2 has a value of 12.9. The worst performing policy is policy 6 with an absolute metric value of 13.8 for the MWL. Additionally, Figure 3 shows that most policies have high metric values for some of the variables and low values for others. In other words, their consistency depends on the outcome observed. Overall, policies 6 and 1 have the worst-performing signal-to-noise ratio values. In five of the six outcomes, policy 6 is among the two policies with highest metric values. Thus, it is consistently one of the most inconsistent policies. The figure of normalized values also shows that policy 9 has close to the lowest normalized values from all the policies. However, in only MWL it is the policy with the lowest metric value. Nevertheless, it can be said to be one of the best performing policies on this metric and in the most consistent policies in this set.

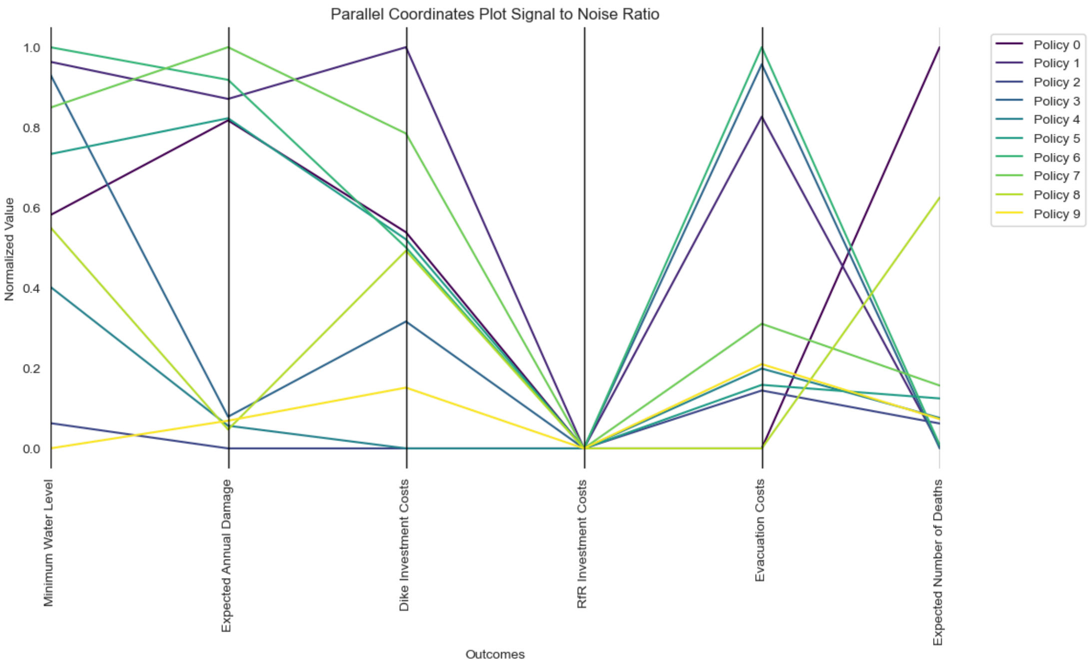


*Figure 3: Parallel Coordinates for the 10 candidate policies and their Signal to Noise ratio for each of the outcomes*

In the maximum regret robustness metric there is more variability compared to the signal-to-noise ratio as can be seen in Figure 4. There is a clear overall best performer: policy 7. It is in the top 5 policies in 5 of the six outcomes. It performs somewhat poorly only in *RfR Investment Costs*, while still not being among the worst performers.

The policies with the lowest maximum regret values in MWL are policy 2 and 9. However, the overall metric performance is poorer compared to policies 7 and 0. The absolute MWL regret values of policies 2, 9, 7, and 0 are 0.59, 0.63, 0.92, and

0.77 meters, respectively. Since MWL is the most important outcome variable for the problem owner, there is a justification in emphasizing it. Thus, policy 0 offers the best balance in the maximum regret performance. It has a lower maximum regret in MWL while having a decent overall performance, being among the worst-performers only in *RfR Investment Costs* and *Evacuation Costs*.

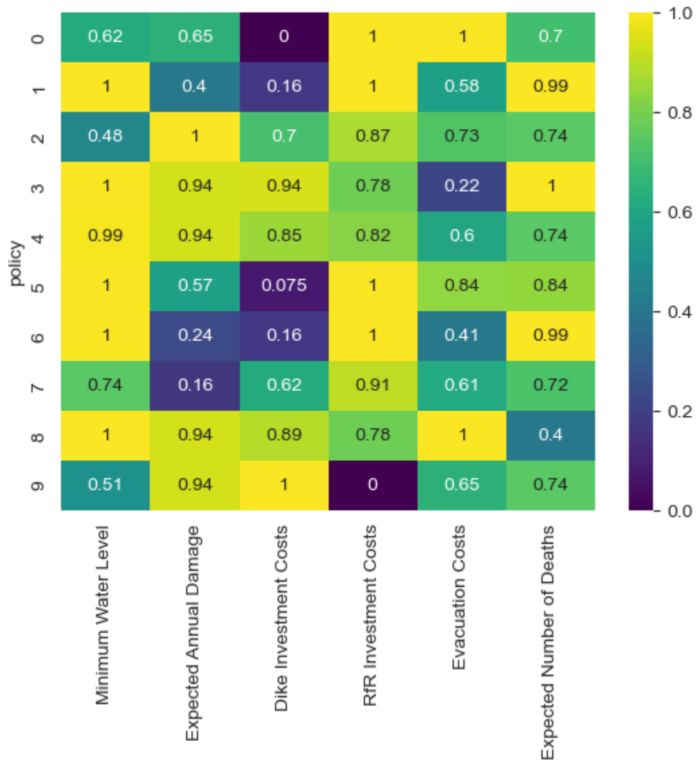


*Figure 4: Maximum Regret Scores*

Looking at both of the robustness metrics, policies 2, 9, and 4 are the overall most robust. The policies 2 and 9 are also the ones which have the best performance in MWL in both metrics. Policy 4 has one of the highest MWL maximum regret and is thus not a good performer when emphasizing the MWL. The identified best overall performing policy in maximum regret: policy 0, has relatively high signal-to-noise ratio values and is thus not a good performer when looking at both of the metrics. Thus, policies 2 and 9 are the most robust policies after considering both robustness metrics and emphasizing the MWL.

### 3.3.3. PRIM

As the last step of MORDM, a second PRIM iteration is made to identify the uncertainties which may cause the obtained policies to perform poorly (Bryant & Lempert, 2010; Appendix E). Since all outcomes exceed the required MWL of 4, a similar target is used as in the reduction of candidate strategies: the outcomes with the 75% highest *Expected Number of Deaths* or *Expected Annual Damage*. This target represents the set of undesirable outcomes. With *peel_alpha* = 0.1, the algorithm ends up restricting only one dimension to achieve density of 1. A box with a coverage of 0.79 and a density of 0.86 is chosen. The identified uncertainty variable is *A.1_pfail*, representing the probability of dike 1 withstanding the loads. Probability values below 0.73 are associated with an undesirable outcome. In other words, a higher probability of withstanding the loads is associated with desirable outcomes and the policy may fail with low probabilities.

### 3.3.4 Policy Recommendations

The results gathered from the MORDM analysis suggest that the most robust policies for the TC that allow an optimal *MWL* while minimizing the *Expected Number of Deaths* and *Expected Annual Damage* are policies 2 and 9. These policies are robust across scenarios and could help our client in advocating for its interests while making alliances with other actors in the decision arena. Table 1 shows the location of the *Room for the River* projects and dike reinforcements, as well as the provinces in which they are situated for each policy.

| Policy | RfR Projects | RfR Province | Dike Increase Locations | Dike Increase Province |
|---|---|---|---|---|
| 2 | RfR 0 Olburgen | Gelderland | A2 Cortenoever | Gelderland |
| | | | A3 Zutphen | |
| | RfR 2 Tichelbeeksewaard | Gelderland | A4 Gorssel | Overijssel |
| | | | A5 Deventer | |
| 9 | RfR 0 Olburgen | Gelderland | A2 Cortenoever | Gelderland |
| | | | A3 Zutphen | |
| | RfR 2 Tichelbeeksewaard | Gelderland | A4 Gorssel | Overijssel |
| | RfR 4 Obstakelverwijdering | Gelderland | A5 Deventer | |

Table 1: Policy Recommendations for Transport Company

# 4. Discussion

This report aimed to use exploratory modelling techniques to find the optimal and most robust policies for the TC to achieve its goals. Several limitations emerged that constrain the validity of the conclusions and point directions for further improvement. These limitations span both technical and political dimensions and are important to acknowledge due to the context for which the model was developed.

Essentially, the original model did not include the *MWL* as an outcome, and had to be manually added to explore the TC's main interest: identify policies that maintain the water level above 4 meters. The results obtained from the open exploration and the directed search remain somewhat disconnected from the debates, since no other actor implemented this outcome into their models. Given the rational, data-driven approach taken by Rijkswaterstaat and Delta Commision, the TC, and by extension us as analysts, struggled to establish the legitimacy of these results and weakened TC's position to demand these desired policies during the debates.

Some key weaknesses of our approach emerged during the open exploration phase. At the beginning of the analysis, we were only focusing on the problem framing of our client, and not so much on the exploration of potential outcomes that could benefit them in terms of alliances. The client's mandate was vague and offered little guidance, leading modelers to define what best means in terms of maximizing trade volume. As a result, we did not fully leverage the potential of PRIM, and the uncertainty space remained limited. Additionally, in all scenarios the *MWL* remained above 4 meters, meaning worse-case scenarios were already acceptable for our client. Knowing this, we opted to further explore a higher minimum threshold for our client by choosing a 5-meter threshold for further analysis, but too few cases met this condition to apply the PRIM algorithm.

Furthermore, the IJssel River Transport Company's goals were sensitive to a number of uncertain parameters due to the company's role on the entire river. The transport company is affected by the minimum water level. This is unlike local actors. The minimum water level must be met at any point along the river. Therefore, most variables impacted the outcomes. This was demonstrated in two ways. Firstly, the Random Trees algorithm showed that the ranking was determined by barely noticeable differences. Secondly, the Sobol analysis revealed that second-order interactions were primarily responsible for the variance in outcomes. Overall, the flood wave shape appeared to have the greatest impact during open exploration. However, we were unable to immediately incorporate the importance of the flood wave shape variable after discovering its relevance, and had to postpone addressing it until the optimisation phase. This phase showed that the variable was not as important for robust policies. This suggests that robust policies (such as 2 and 9) may be able to overcome the influence of the flood wave shape despite its relevance in the general solution space. Nevertheless, further research is needed to better understand the role of this variable.  were only able to incorporate the importance of the *Flood Wave Shape* variable immediately after discovering its relevance, and had to postpone addressing it until the optimization phase.

The MORDM framework implemented for the directed search was built based on one single reference scenario obtained with the PRIM algorithm. The use of a reference scenario is a recognized weakness of MORDM. The candidate solutions derived from this approach might perform poorly in future states of the world that deviate from the baseline used during the search (Bartholomew & Kwakkel, 2020). Furthermore, the selection of this specific reference scenario could be enhanced. The current reference scenario comes from one of the PRIM boxes. Alternatively, interesting reference scenarios could have come using an optimization-based selection procedure which optimises the diversity of scenarios (Eker & Kwakkel, 2018).

One of the major constraints we faced was the lack of computational power to run multiple seeds and scenarios with MOEA. Although we attempted to use the Delft Blue Supercomputer, our job remained in the queue for several days without progress. As previously noted, more reference scenarios were needed, and additional seeds could have reduced the model's stochastic uncertainty and improved the overall Pareto front.

The lack of guidance from our client continued to play a role in this part of the analysis, as the pareto optimal confirmed what we already intuit: *MWL* was above 4 meters always. It became crucial to address other framings to get an understanding of the multi-actor environment. Hence, the *Expected Number of Deaths* and E*xpected Annual Damage were added* as outcomes of interest to study potential alliances with actors like Delta Commision. As for future improvement, a takeaway is addressing the multi-actor environment by adding foreign interest of outcomes should be done earlier in the analysis.

One of the political drawbacks of these policy recommendations is that all *Room for the River* projects take place in Gelderland. Indeed, this proposal leads to a desirable trade-off between safety and MWL. Nevertheless, the TC is aware of Gelderland opposition to RfR in its region due to their reliance on agriculture. As it happened in the debate, it is expected that the region will oppose it. Therefore, in future research other metrics should be added to further analyze the robustness of the candidate policies in a truly multi-actor context.

Moreover, these policies do not equally distribute its burdens, leaving Gelderland with most of the large interventions. In this report, a utilitarian approach has been pursued. In future research, perhaps an egalitarian one should be analysed, aiming to find a compromise between effectiveness, safety and justice.

The exploration of the *A.0_ID flood shape* uncertainty parameter could be improved in future research. Multiple values were tested in the reference scenario, during the optimization and the analysis part of MORDM. However, in the rest of the algorithm only a fixed value of *A.0_ID flood shape* to computational power constraints. As a result, no definitive conclusions can be drawn about the sensitivity of outcomes to this uncertainty.

Eight different A.0_ID flood values in the reference scenario were used to explore the sensitivity of outcomes to this uncertainty parameter, which had not been analyzed in earlier stages of the report. Variations in this parameter did not result in any scenario with a MWL below 4 meters. Moreover, the plots in Appendix C indicate that trade-offs behave similarly across simulations. For example, a higher MWL generally leads to an increase in the *Expected Number of Deaths*. However, there are differences in the outcomes values and the selected candidate policies.

Finally, additional iterations using the last results from PRIM could provide a new reference scenario that could serve to perform a second iteration of MORDM to ensure the robustness of the policies we recommended to our client. The latter issues could have been improved with more iterations of MORDM, with clearer guidance from TC (more reference scenarios to explore different outcomes).

# 5. Conclusions

## 5.1. Exploration Objectives

The exploration successfully found robust policies to ensure a *MWL* for the transport company while also taking into account the views of other stakeholders. The analysis found that the MWL of 4 meters is expected in all scenarios, thus any policy satisfies the minimum needs of the Transport Company (TC). Since no scenario was found with MWL below 4 meters, the objective of the analysis shifted towards improving the water level and reducing undesired outcomes such as damage and deaths. By doing so, the resulting policies are most robust not only to uncertainties but also initially rival problem framings.

The identified policies ensure the relevance of the TC in the decision-making since the focus became broader than strict adherence to a minimum water level value. Nevertheless, the policies identified do not compromise the core interests of the TC in ensuring a navigable minimum water level. It is assumed that a higher minimum water level value is preferred by the TC, even after the set requirement is satisfied. Thus, after satisfying the primary objective, a secondary objective became to find the most advantageous policies for the TC while taking into account broader concerns. In other words, the objective was to aim to maximize the MWL while lowering the *Expected Annual Damage* and *Expected Number of Deaths*.

There are trade-offs between the different outcome variables. A higher MWL tends to increase the *Expected Number of Deaths* and *Expected Annual Damage*. Thus, instead of solely maximizing the MWL, the objective was to minimize the undesired outcomes while ensuring a satisfactory MWL.

## 5.2. Key Variables

### 5.2.1. Key Policy Levers

The key policy levers are the RfR project Welsummer Buitenwaarden and dike increase in Deventer which are influential for the MWL and the *Expected Number of Deaths*, respectively. The dike increase in Doesburg is the most influential for *Expected Annual Damage* and also plays a role on the MWL. The upstream locations such as Welsummer Buitenwaarden and Doesburg have a high influence in the model since their effects are experienced throughout the river, also in the downstream areas. A major population center along the river, Deventer, has a high influence due to the number of residents and thus the potential number of people affected by the river.

### 5.2.2. Key Uncertainties

The key uncertainties affecting the MWL are the flood wave shape, probability of dike 1 withstanding loads and maximum breach width of dike 2. Out of these, the flood wave shape is a special case since it is a categorical variable. Though flood wave shape affects the MWL, the outcomes of the identified robust policies are not particularly sensitive to it. Probability of dike 1 withstanding loads is influential because of the dike's upstream location. The maximum breach width of dike 2 has notable second-order interactions. The identified robust policies may perform poorly if the probability of dike 1 withstanding the loads is lower.

## 5.3. Policy Recommendation

### 5.3.1. Identified Policies

Two robust policies were identified. The policies are presented in Table 1 in Section 3.3.4. Both of the policies are a combination of RfR projects and dike increases. Namely, RfR projects Olburgen (RfR 0) and Tichelbeeksewaard (RfR 2) along with dike increases in Cortenoever (A2), Zutphen (A3), Gorssel (A4), and Deventer (A5). Additionally policy 9 includes RfR project Obstakelverwijdering (RfR 4) which is not included in policy 2. The two policies differ in their dike increase heights and implementation time steps. Overall the analysis shows that instead of an either-or choice between RfR projects and dike increases, it is in the interests of the TC to have both types of policies implemented. The locations and specifics such as dike increase heights play an important role in ensuring a preferred outcome for the TC. The details of the recommended policies 2 and 9, as well as other identified policies can be found in Appendix D. The identification of these policies included other outcomes in addition to the minimum water level and thus also took into account other stakeholders interests.

### 5.3.2. Policy Robustness

Two robustness metrics, signal-to-noise ratio and maximum regret, are used to evaluate the robustness of the policies. Policies 2 and 9 have low values in both metrics, especially for the MWL but also in other outcome variables and are thus the most robust policies satisfying the objectives of the exploration. Some alternative policies showed higher overall robustness in the maximum regret metric. However, the MWL was emphasized against other outcome variables when choosing the most robust policies. Furthermore, the maximum regret has a high level of aggregation and high intrinsic risk aversion. Since the objective is not to solely minimize the *Expected Annual Damage* or any other outcome variable, a deviation from the most conservative approach can be justified. Both policies — 2 and 9 — have relatively low scenario outcome dispersion, shown by the low normalized signal-to-noise ratio values. It is assumed that a consistent policy is preferred by the TC as well as other stakeholders.

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# Appendix

## Appendix A:

| **Uncertainties** | Locations | Range [unit] | Explanation |
|---|---|---|---|
| discount rate | 0, 1, 2 (timestep) | 1.5, 2.5, 3.5, 4.5 [dmnl] | discount rate for calculating present day value of damages |
| pfail | A.1 - A.5 | 0 - 1 [dmnl] | probability that dike will withstand the hydraulic load |
| Bmax | A.1 - A.5 | 30 - 350 [metres] | final extent of breach width |
| Brate | A.1 - A.5 | 0, 1.5, 10 [1/day] | how fast the breach grows over time |
| A.0_ID flood wave shape | - | 0 - 132 [dmnl] | normalized curve describing the shape of incoming flood wave over time |
| **Outcomes** | | | |
| Minimum Water Level | - | [metres] | the minimum water depth |
| Expected Annual Damage | - | [€] | discounted expected annual flood damage |
| Dike Investment Costs | - | [€] | investment costs of dike raising |
| RfR Investment Costs | - | [€] | investment costs for room for the river projects |
| Evacuation costs | - | [€] | costs of evacuation based on number of |

| | | | |
|---|---|---|---|
| | | | people and the duration they have to leave their home |
| Expected Number of Deaths | - | [person] | expected annual number of casualties due to floods |
| **Levers** | | | |
| RfR | 0 - 4 | 0 / 1 [dmnl] | whether to activate the RfR project at the specified location or not. |
| DikeIncrease | A.1 - A.5 | 0-10 [decimeter] | amount of dike raising |
| EWS_DaysToThreat | - | 0-4 [days] | number of days prior to threat to give a warning |

*Table A.1 : uncertainties, outcomes and levers of model with explanation*

The model simulates over 200 years, this was also decided upon during the first debate. The levers RfR and Dike increase therefore also have a time dimension. They can either be implemented at timestep 0, at the beginning, timestep 1 after 66 years and timestep 2 after 133 years. The locations of the RfR projects and the dike rings are shown in Figure A.2.


*Figure A.2 : Room for river locations and dike ring locations (Topomania, n.d.)*

## Appendix B:

The results of the open exploration of the scenario space (B.1. and B.2.) can be found in notebook "Open_exploration_scenario_policy.ipynb" and is available in the GitHub repository (see link in Appendix F).

## B.1. Random Trees

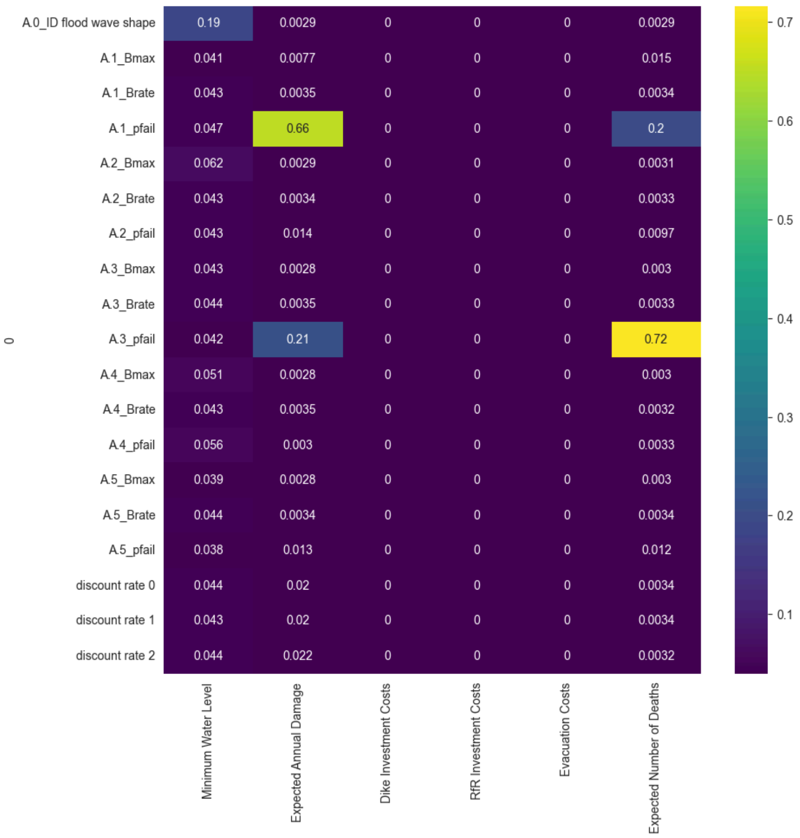


*Figure B.1 : Random Trees*

## B.2. Reference Scenario

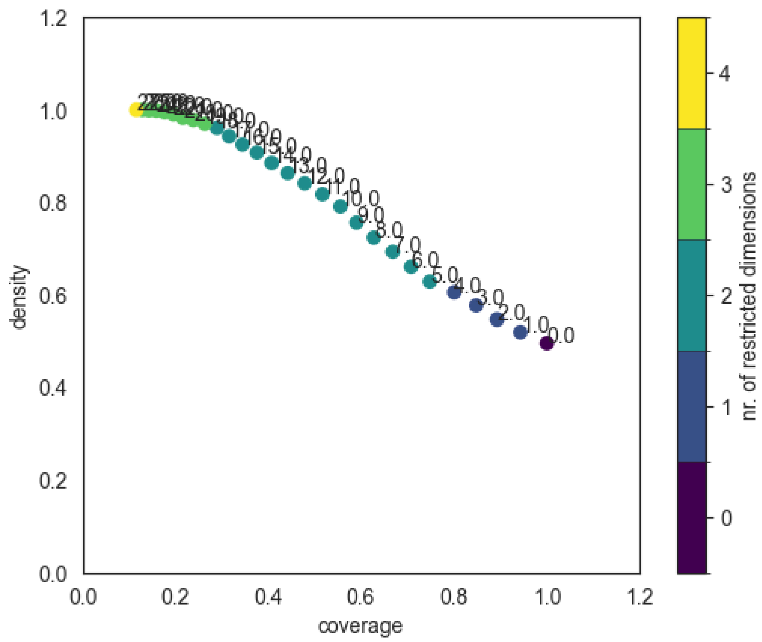


*Figure B.2 : PRIM trade-off curve made from SOBOL results*

From this PRIM outcome, box number 8 is chosen to create the reference scenario. Box 8 is chosen for its balance between coverage and density. The boundary data of this box is saved and the midpoint values are calculated to get the reference scenario, that has the following parameters: "Scenario({'discount rate 0': 1.5, 'discount rate 1': 1.5, 'discount rate 2': 1.5, 'A.0_ID flood wave shape': 30, 'A.1_Bmax': 190.0, 'A.1_pfail': 0.6720691719092429, 'A.1_Brate': 1.0, 'A.2_Bmax': 245.05131609737873, 'A.2_pfail': 0.5, 'A.2_Brate': 1.0, 'A.3_Bmax': 190.0, 'A.3_pfail': 0.5, 'A.3_Brate': 1.0, 'A.4_Bmax': 190.0, 'A.4_pfail': 0.5, 'A.4_Brate': 1.0, 'A.5_Bmax': 190.0, 'A.5_pfail': 0.5, 'A.5_Brate': 1.0})"

## B.3. Random policies

The results of the open exploration of the policy space (B.3. and B.4.) can be found in notebook “Open_exploration_scenario_policy.ipynb” and is available in the GitHub repository (see link in Appendix F).

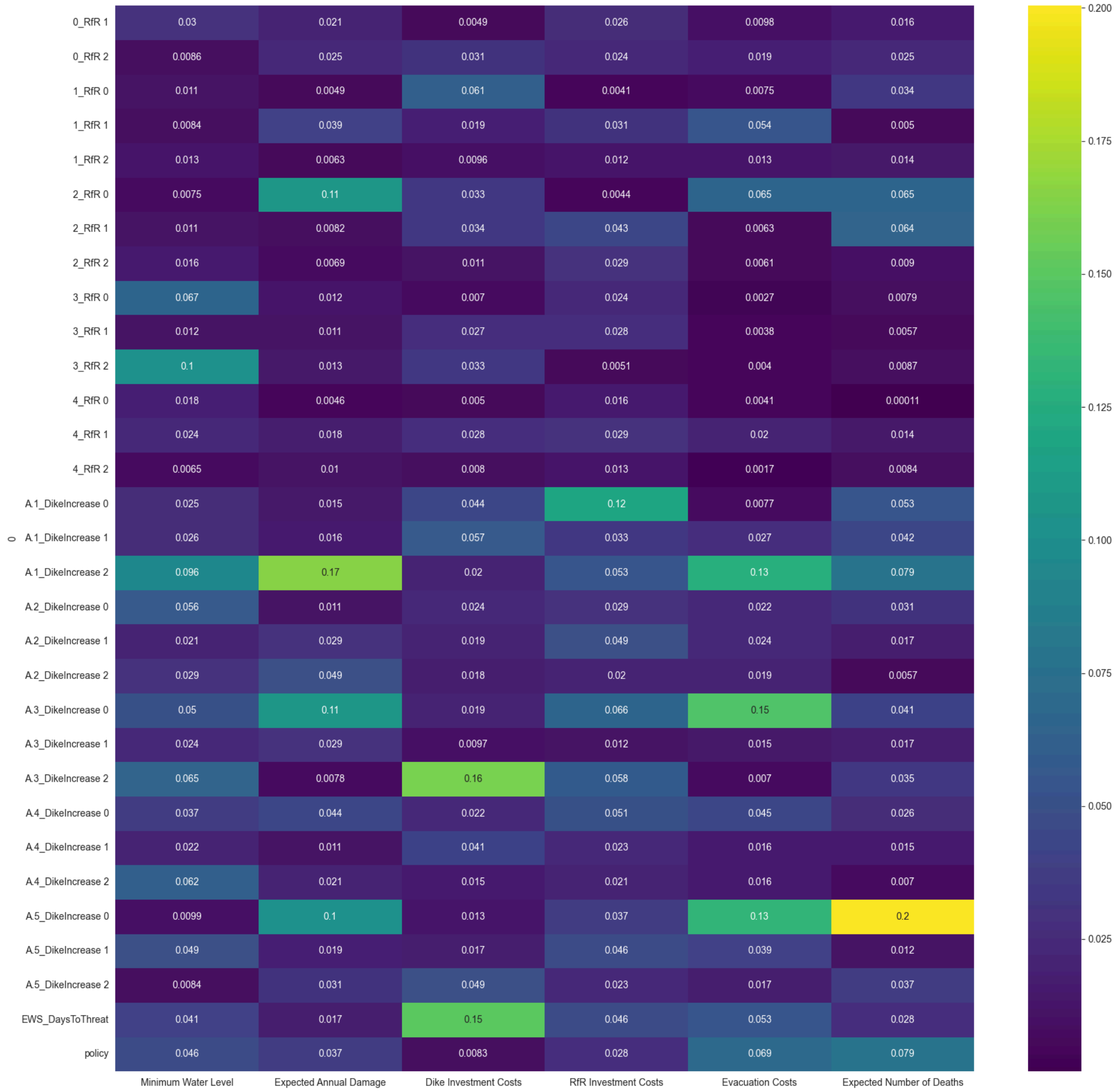


*Figure B.3.1.: Heatmap of scores of random policies*

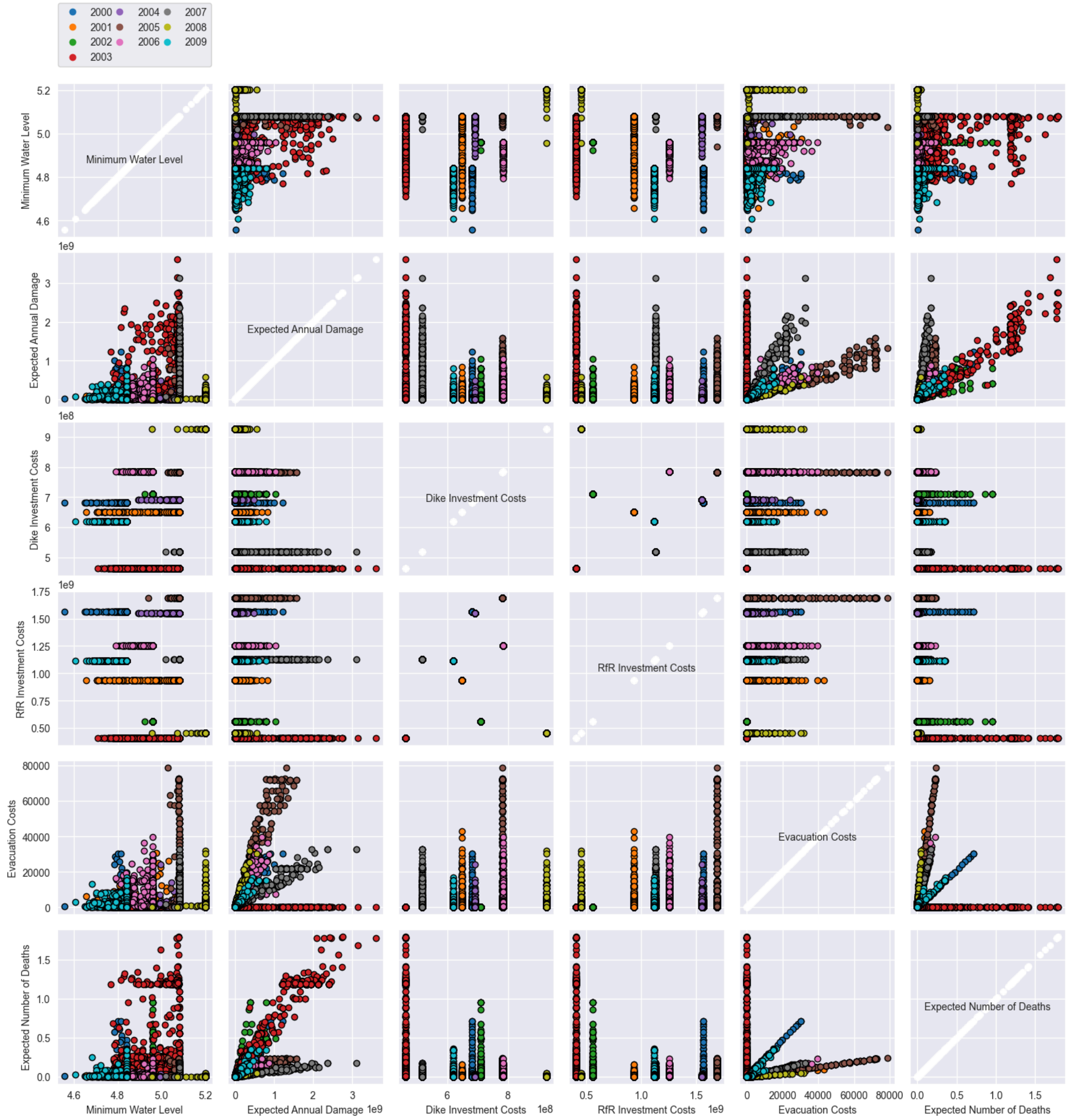


*Figure B.3.2.: Pairplot of results of random policies*

## B.4. Debate-informed Policies

| Policy | Levers |
|---|---|
| P0 BAU | All levers = 0 |
| P1 | • 0_RfR 0 : 1<br>• A.1_DikeIncrease 0 : 5 |
| P2 | • 4_RfR 0 : 1 |

| | |
|---|---|
| | • A.5_DikeIncrease 0 : 5 |
| P3 | • 1_RfR 0: 1<br>• A.3_DikeIncrease 0 : 5 |
| P4 | • 1_RfR 0 : 1<br>• 2_RfR 0 : 1 |
| P5 | • 2_RfR 0 : 1 |
| P6 | • 1_RfR 0 : 1 |
| P7 | • 1_RfR 0 : 1<br>• 2_RfR 0 : 1<br>• 2_DikeIncrease 0 : 5<br>• 4_DikeIncrease 0: 5 |
| P8 | • 1_RfR 1 : 1<br>• 2_RfR 1 : 1 |
| P9 | • 1_RfR 1 : 1<br>• 2_RfR 1 : 1<br>• 2_DikeIncrease 1 : 5<br>• 4_DikeIncrease 0 : 5 |
| P10 | • 1_RfR 0 : 1<br>• 3_RfR 0 : 1<br>• 2_DikeIncrease 0 : 5<br>• 4_DikeIncrease 0 : 5 |
| P11 | • 1_RfR 0 : 1<br>• 3_RfR 0 : 1 |
| P12 | • 0_RfR 0 : 1<br>• 1_RfR 0 : 1<br>• 2_RfR 0 : 1<br>• 3_RfR 0 : 1<br>• 4_RfR 0 : 1 |
| P13 | • 0_RfR 0 : 1<br>• 2_RfR 0 : 1<br>• 1_DikeIncrease 0 : 5<br>• 4_DikeIncrease 0 : 7<br>• 5_DikeIncrease 0 : 8 |

*Table B.4 : Selected policies informed by discussions and debates*

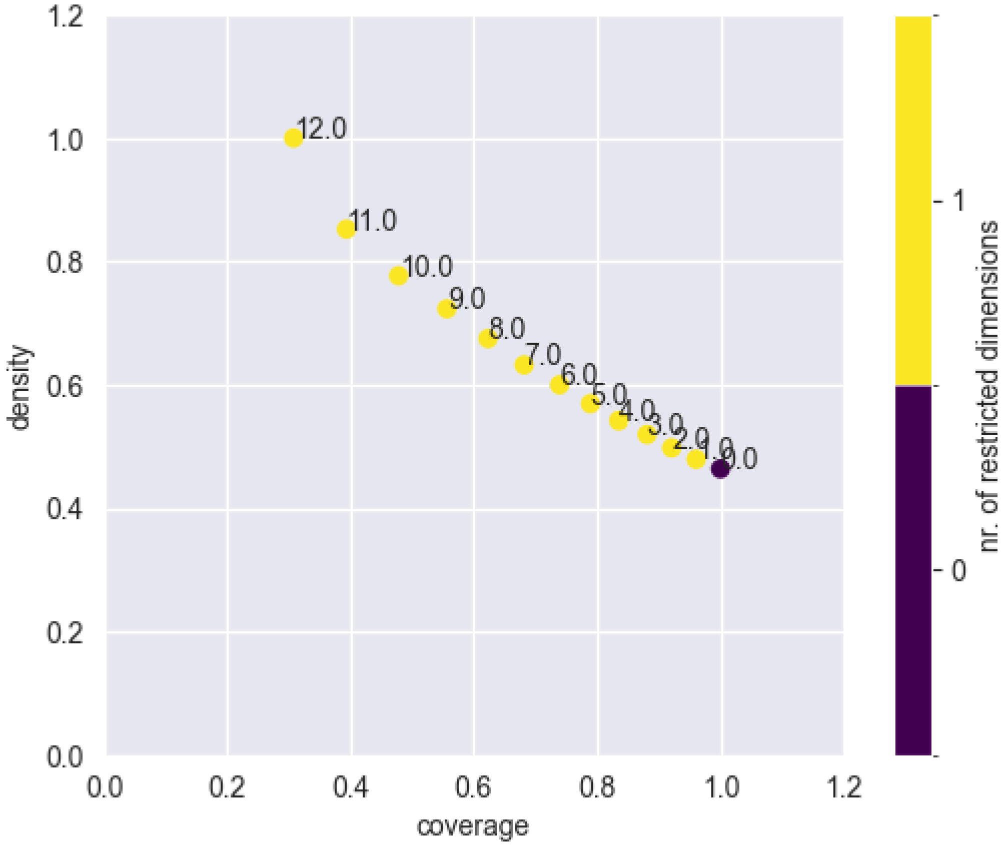


Figure B.4.1 : PRIM trade-off curve performed on selected policies

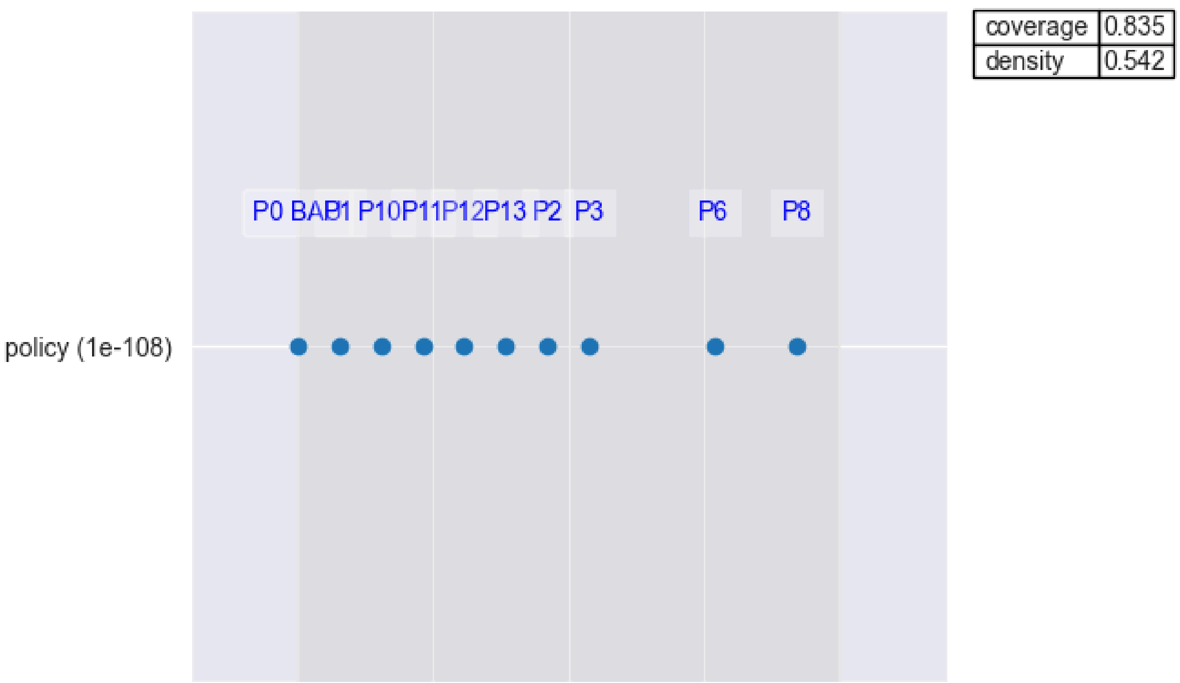


*Figure B.4.2 : PRIM outcomes of chosen box*

## Appendix C: Trade-Offs for Candidate Strategies with different Flood Shape

The plots were generated using the notebooks titled *MORDM Analysis fw_shape_seed.ipynb* (one file per seed, e.g., *MORDM Analysis fw_shape_22.ipynb*) and are available in the GitHub repository (see link in Appendix F)."

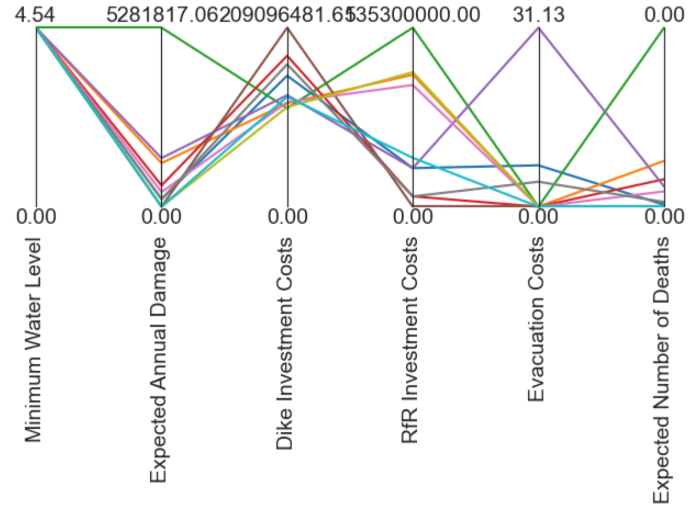


*Figure C.1: Flood Shape 0*

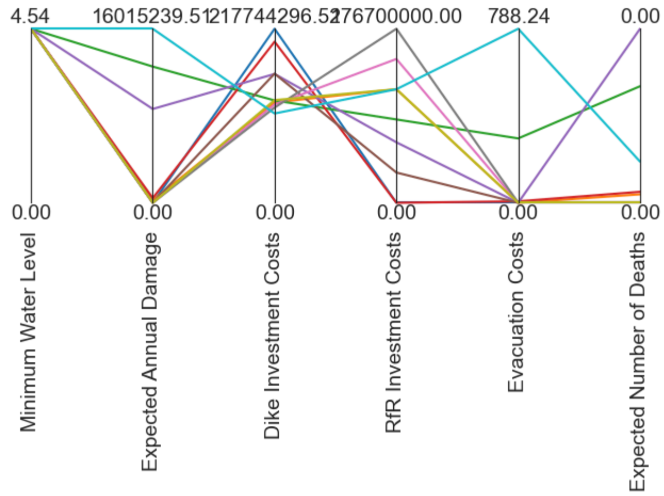


*Figure C.2: Flood Shape 22*

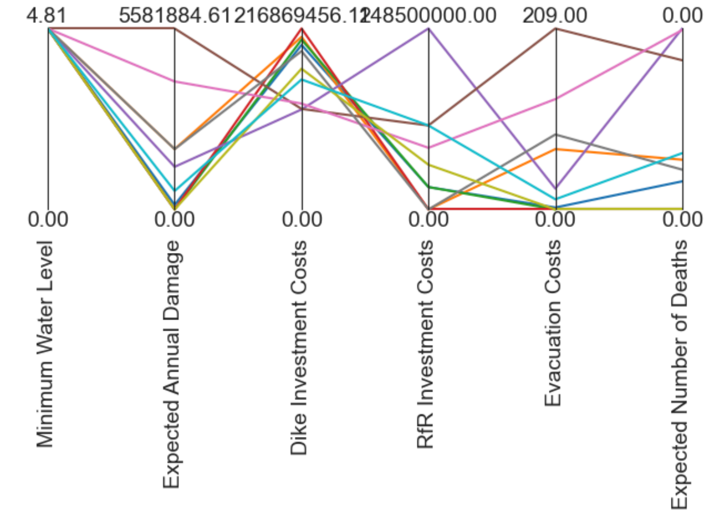


*Figure C.3: Flood Shape 44*

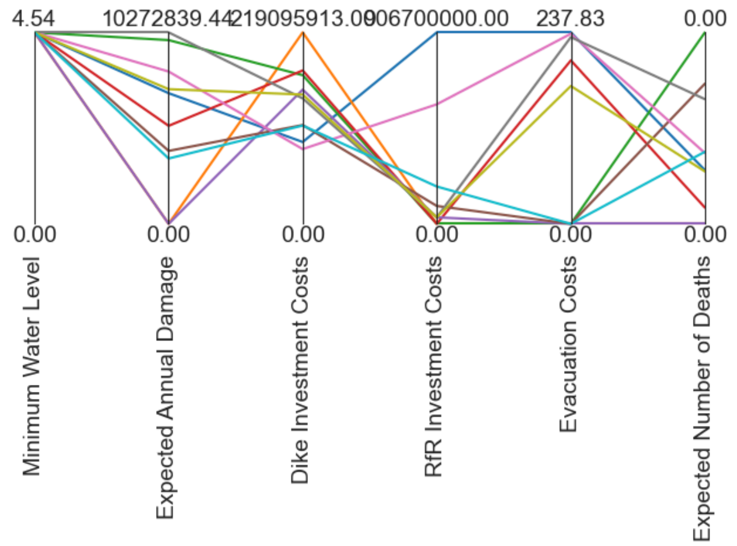


*Figure C.4: Flood Shape 66*

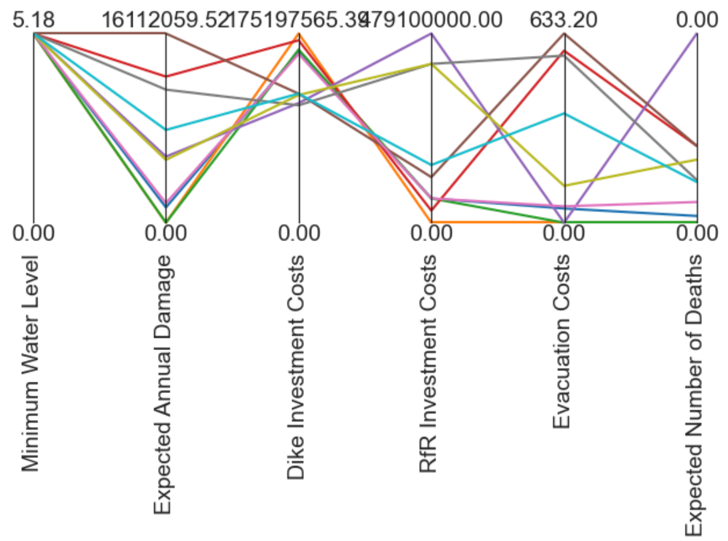

*Figure C.5: Flood Shape 88*

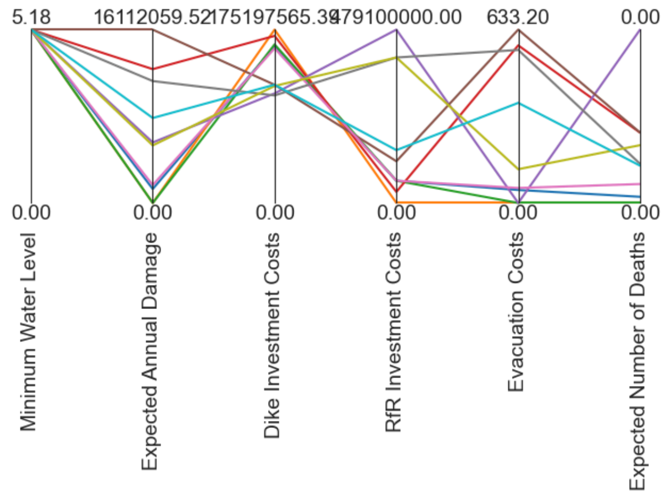


*Figure C.6: Flood Shape 110*

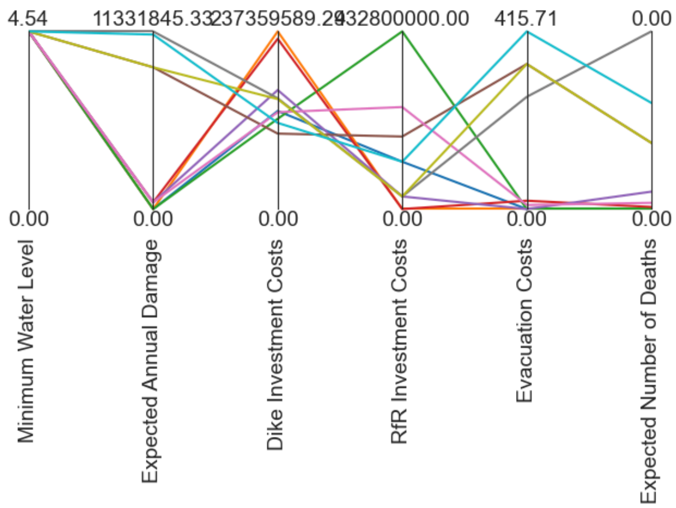


*Figure C.7: Flood Shape 132*

# Appendix D. Identified Policies

Table D.1. Identified Pareto optimal policies including the recommended policies 2 and 9.

| Policy | Time Step | 0 | 1 | **2** | 3 | 4 | 5 | 6 | 7 | 8 | **9** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RfR_0 Olbrugen | 0 | 0 | 0 | **1** | **1** | 0 | 0 | 0 | 0 | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | **1** | 0 | 0 | 0 | 0 | **1** |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **1** | 0 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RfR_1 Havikerwaard | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| RfR_2 Tichelbeeksewaard | 0 | **1** | 0 | 0 | **1** | 0 | **1** | 0 | **1** | **1** | **1** |
| | 1 | 0 | **1** | **1** | **1** | **1** | 0 | **1** | **1** | **1** | **1** |
| | 2 | 0 | 0 | 0 | **1** | **1** | 0 | 0 | **1** | **1** | **1** |
| RfR_3 Welsummer Buitenwaarden | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| RfR_4 Obstakelverwijdering | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **1** |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **1** |
| Days to Threat | | 0 | **3** | **1** | **3** | **1** | **1** | **3** | **1** | 0 | **1** |
| Dike Increase A1 Doesburg | 0 | **1** | **1** | 0 | 0 | 0 | **1** | **1** | **1** | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Dike Increase A2 Cortenoever | 0 | **4** | **4** | **3** | **3** | **3** | **6** | **4** | **1** | **3** | **1** |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Dike Increase A3 Zutphen | 0 | **6** | **4** | **4** | **1** | **2** | **5** | **3** | **1** | **1** | **1** |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Dike Increase A4 Gorssel | 0 | **4** | **3** | **6** | **1** | **3** | **3** | **3** | **3** | **2** | **3** |
| | 1 | **1** | 0 | 0 | 0 | 0 | 0 | **1** | 0 | 0 | 0 |
| | 2 | **1** | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Dike Increase A5 Devented | 0 | **5** | **5** | **5** | **3** | **4** | **5** | **6** | **3** | **4** | **4** |
| | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

## Appendix E. Second PRIM Output Graphs

The plots were generated using the notebook titled *MORDM PRIM.ipynb* and is available in the GitHub repository (see link in Appendix F).

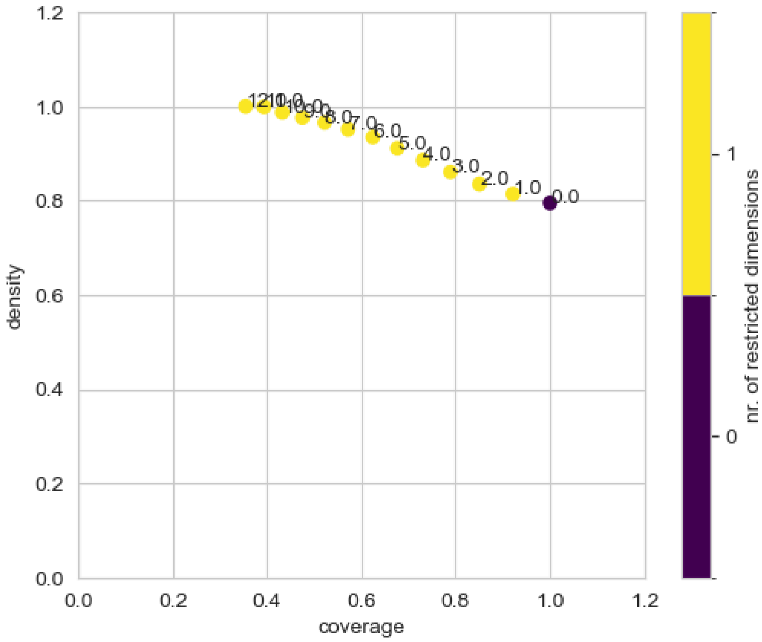


Figure E.1. PRIM trade-off curve performed on robust policies

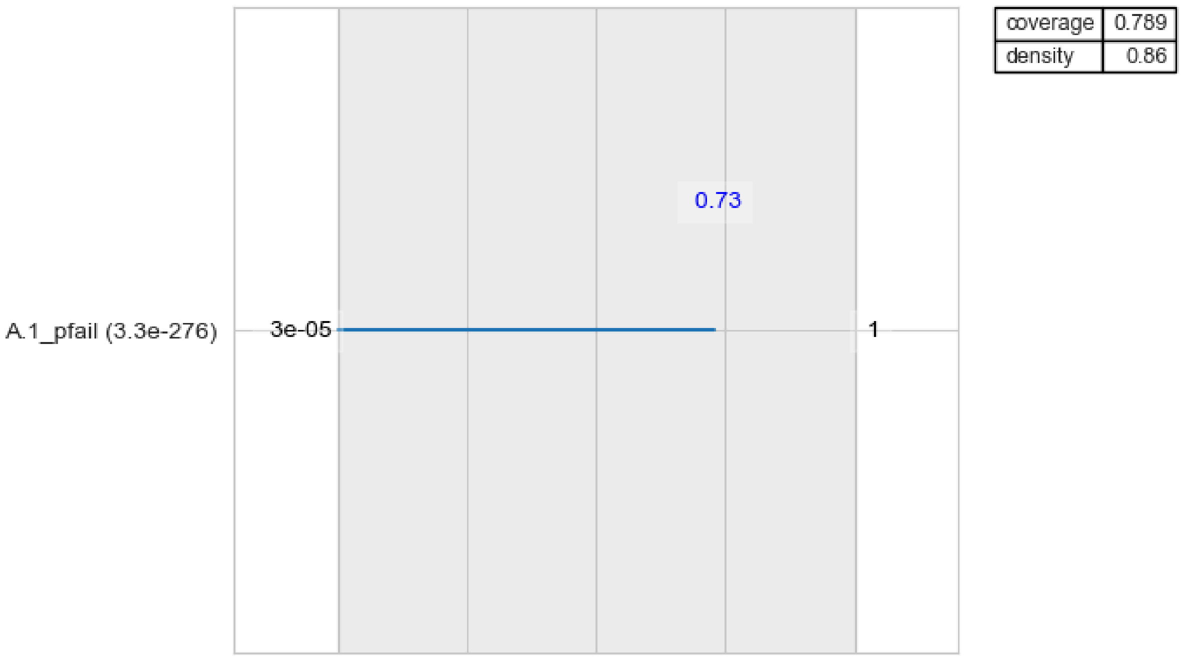


Figure E.2. *PRIM outcomes of chosen box*

## Appendix F. Github Repository

The code containing the model and the methods for analysis and plots used in this report can be found in the next link in the repository *MBDM-Group-1*.
https://github.com/daniela1998/MBDM-Group-1.git